

Project Overview

Parameterizable Multi-Port Memory Subsystem
using Round Robin Arbitration

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1 Project Title

Design and Implementation of a Parameterizable Multi-Port Memory Subsystem using Round Robin Arbitration

2 Project Objective

The objective of this project is to design and implement a scalable and synthesizable multi-port memory subsystem capable of handling simultaneous requests from multiple clients. The design ensures fair access to shared memory resources through a Round Robin Arbitration scheme while maintaining modularity and parameterization.

The project demonstrates concepts frequently used in modern SoC, NoC, DMA, cache, and memory subsystem architectures.

3 Problem Statement

In complex digital systems, multiple processing elements often require access to a shared memory resource. Direct access may lead to:

- Resource conflicts
- Starvation of lower priority clients
- Increased system complexity
- Reduced throughput

Hence, an efficient memory subsystem with arbitration and transaction management is required.

4 System Architecture

The complete subsystem consists of the following modules:

1. Round Robin Arbiter
2. Request Multiplexer
3. Memory Controller FSM
4. Dual-Port RAM
5. Response Demultiplexer
6. Performance Monitoring Counters

Architecture Flow

Client Requests → Arbiter → Request MUX → Memory Controller → Dual-Port RAM
→ Response DEMUX

5 Major Features

- Supports multiple clients accessing shared memory.
- Fair arbitration using Round Robin scheduling.
- Parameterized data width and address width.
- Modular RTL design.
- Synthesizable on FPGA platforms.
- Built-in performance counters.
- Scalable architecture for future enhancements.

6 Modules Description

6.1 Round Robin Arbiter

Provides fair memory access by rotating priorities among requesting clients and preventing starvation.

6.2 Request Multiplexer

Selects the address, data, and control signals of the granted client.

6.3 Memory Controller

Controls memory transactions using an FSM consisting of:

- IDLE
- PROCESS
- COMPLETE

6.4 Dual-Port RAM

Stores memory contents and provides independent memory access ports.

6.5 Response Demultiplexer

Routes read data and ready signals back to the requesting client.

6.6 Performance Counters

Maintains statistics regarding:

- Read transactions
- Write transactions
- Grants issued
- Total transactions

7 Design Parameters

```
parameter DATA_WIDTH = 8;  
parameter ADDR_WIDTH = 4;
```

The parameterized design allows easy scalability without major RTL modifications.

8 Simulation and Verification

Functional verification was carried out using Vivado Simulator.

Verification included:

- Multiple client requests
- Read transactions
- Write transactions
- Arbitration behavior
- Counter verification
- Waveform analysis

9 Synthesis Results

The design was synthesized successfully using Xilinx Vivado.

- Slice LUTs Used : 69
- Slice Registers Used : 154
- Timing Violations : None
- Modular Hierarchy Preserved

The subsystem occupies less than 1% of available FPGA resources, demonstrating efficient RTL implementation.

10 Applications

Possible applications include:

- Network-on-Chip (NoC)
- DMA Controllers
- Shared Cache Systems
- Embedded SoCs
- Multi-core Processors
- AI Accelerator Memory Subsystems

11 Future Scope

Future enhancements may include:

- Support for 8/16 clients
- Burst transactions
- QoS-based arbitration
- ECC-protected memory
- Pipelined controller
- Cache integration
- AXI interface support

12 Skills Demonstrated

This project demonstrates knowledge in:

- Verilog HDL
- RTL Design
- FSM Design
- FPGA Design Flow
- Arbitration Algorithms
- Memory Architectures
- Functional Verification
- Synthesis and Timing Analysis
- Performance Monitoring

13 Conclusion

A parameterizable multi-port memory subsystem was successfully designed and synthesized. The project demonstrates important digital design concepts including arbitration, shared memory management, FSM-based control, modular RTL design, and scalability.

The architecture serves as a strong foundation for advanced FPGA and ASIC memory subsystem designs.

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